

# Types of Variables

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## Types of variables

Before summarizing data, it helps to identify the **type** of each variable, since the appropriate descriptive methods depend on the type. Variables are broadly classified as either **numerical** or **categorical**, and within each class there are important subtypes.

**Definition 0.1** (Numerical variable). A **numerical** (or **quantitative**) variable is a variable that takes values on a numeric scale, where arithmetic operations such as subtraction (and possibly division) are meaningful. Numerical variables may be further classified as **interval** or **ratio** variables, and as **continuous** or **discrete**.

**Example 0.1** (Numerical variables in the WCGS). Age, systolic blood pressure, total cholesterol, and number of cigarettes smoked per day are numerical variables in the WCGS dataset.

**Definition 0.2** (Interval variable). An **interval variable** is a numerical variable for which differences between values are meaningful, but there is no natural zero point (so ratios of values are not meaningful).

**Example 0.2** (Temperature in degrees Celsius). Temperature in degrees Celsius is an interval variable: a difference of 10 deg C is meaningful, but 30 deg C is not “twice as hot” as 15 deg C because 0 deg C is an arbitrary zero point (the freezing point of water) rather than the complete absence of heat.

**Definition 0.3** (Ratio variable). A **ratio variable** is a numerical variable with a natural zero point, so that both differences and ratios of values are meaningful.

**Example 0.3** (Ratio variables in the WCGS). Age, weight, cholesterol, blood pressure, and number of cigarettes per day are ratio variables: a value of 0 represents the complete absence of the quantity, so ratios of values are meaningful.

**Definition 0.4** (Continuous variable). A **continuous variable** is a numerical variable whose possible values form an interval (or union of intervals) of real numbers. A continuous variable is always numerical.

**Example 0.4** (Continuous variables in the WCGS). Age, systolic blood pressure, total cholesterol, and body mass index (BMI) are continuous variables: their possible values form a continuum of real numbers.

**Definition 0.5** (Discrete variable). A **discrete variable** is a variable whose possible values form a countable set. Discrete variables include both **numerical** types (such as [count variables](#)) and **categorical** types (such as binary, nominal, and ordinal variables). In contrast, continuous variables are always numerical.

**Example 0.5** (Discrete variables in the WCGS). In the WCGS dataset:

- number of cigarettes per day is a discrete numerical variable (count);
- coronary heart disease event status is a discrete categorical variable (binary).

**Definition 0.6** (Categorical variable). A **categorical** (or **qualitative**) variable is a variable that takes values in a finite set of categories, where arithmetic operations such as differences are not meaningful. Categorical variables may be **nominal** (unordered) or **ordinal** (ordered), and are always discrete.

**Example 0.6** (Categorical variables in the WCGS). Behavioral pattern (Type A1, A2, B3, B4), smoking status (yes/no), and weight category are categorical variables in the WCGS dataset.

**Definition 0.7** (Nominal variable). A **nominal variable** is a categorical variable whose categories have no natural ordering.

**Example 0.7** (Nominal variables in the WCGS). Behavioral pattern (Type A1, A2, B3, B4), arcus senilis (present/absent), and ABO blood type are nominal variables: there is no natural numerical ordering among the categories.

**Definition 0.8** (Ordinal variable). An **ordinal variable** is a categorical variable whose categories have a natural ordering.

**Example 0.8** (Ordinal variables in the WCGS). Weight category and age group are ordinal variables: their categories have a natural ordering, though the distances between categories are not measured on a numeric scale.

**Definition 0.9** (Binary variable). A **binary variable** takes only two possible values, often coded 0 (absence) and 1 (presence). A binary variable is a special case of a nominal variable.

**Example 0.9** (Binary variables in the WCGS). Coronary heart disease event status (`chd69`, yes/no) and current smoking status (`smoke`, yes/no) are binary variables.

## Coding a categorical variable with numbers

**Exercise 0.1** (Dummy coding). The Western Collaborative Group Study (WCGS) variable `behpat` (behavioral pattern) has four categories: A1, A2, B3, and B4.

- (a) How many indicator variables are needed to represent `behpat` if A1 is the reference level?
- (b) Write the values of those indicator variables, in the order A2, B3, B4, for a person in category B3, and for a person in category A1.

Solution

*Solution 0.1.*

- (a) 3 indicator variables, one for each non-reference category: A2, B3, and B4.
- (b) In the order (A2, B3, B4): for B3, the values are (0, 1, 0); for A1, the values are (0, 0, 0).

The reference level is the case where all indicators are 0.

**Definition 0.10** (Dummy variables). **Dummy variables** are numeric variables that, together, are a numeric representation of a categorical variable.

Source

Dobson and Barnett (2018, sec. 2.4).

**Example 0.10** (Dummy variables for chromosomal sex). Let  $S$  be the chromosomal sex of a newborn, a categorical variable with the categories “female” and “male”. Let  $M = 1$  if  $S$  is “male” and  $M = 0$  if  $S$  is “female”, and let  $F = 1$  if  $S$  is “female” and  $F = 0$  if  $S$  is “male”. Together,  $M$  and  $F$  are dummy variables for  $S$ .

*Remark 0.1* (Other ways to code dummy variables). There are other ways to construct dummy variables. One is to use the values  $-1$  and  $1$ : for example,  $-1$  for “female” and  $1$  for “male”. See Dobson and Barnett (2018), section 2.4, for details.

**Definition 0.11** (Indicator variable). An **indicator variable** is a [dummy variable](#) whose only values are 0 and 1.

**Source**

Dobson and Barnett (2018, sec. 2.4).

**Example 0.11** (Indicator variables for chromosomal sex). In [Example 0.10](#),  $M$  and  $F$  are indicator variables: each takes only the values 0 and 1. A dummy variable that uses the values  $-1$  and  $1$  ([Remark 0.1](#)) is not an indicator variable.

**Definition 0.12** (Reference level). The **reference level** of a categorical variable is the category that does not have its own [indicator variable](#). An observation in the reference level has the value 0 for every indicator variable of that categorical variable.

**Source**

Dobson and Barnett (2018, sec. 2.4), who call this the corner point parameterization.

**Example 0.12** (The reference level for chromosomal sex). Take chromosomal sex  $S$  with the indicator variable  $M$  for “male” ([Example 0.10](#)). The category “female” has no indicator variable, so “female” is the reference level, and a female newborn has  $M = 0$ . In [Exercise 0.1](#), category A1 of `behpat` is the reference level.

*Remark 0.2* (Any category can be the reference level). The choice of reference level changes how we code the categories. It does not change what the data say. For example, in [Example 0.12](#), we could instead use “male” as the reference level and the indicator

variable  $F$  for “female”. Both choices describe the same two groups.

## Taxonomy of variable types

Figure 1 illustrates the relationships among these variable types.

### Note

The continuous/discrete distinction cuts across the numerical/categorical distinction. Continuous variables are always numerical. Discrete variables include both numerical types (such as count variables) and categorical types (such as binary, nominal, and ordinal variables).

## Variables in the WCGS dataset

Table 1 shows selected variables from the Western Collaborative Group Study (WCGS) dataset and their types.

```
tibble::tribble(
  ~Variable, ~Description, ~Type, ~Scale,
  `age`, "Age (years)", "Continuous", "Ratio",
  `chol`, "Total cholesterol", "Continuous", "Ratio",
  `sbp`, "Systolic blood pressure", "Continuous", "Ratio",
  `bmi`, "Body mass index (kg/m^2)", "Continuous", "Ratio",
  `weight`, "Weight (lbs)", "Continuous", "Ratio",
  `ncigs`, "Cigarettes per day", "Count (discrete)", "Ratio",
  `chd69`, "CHD event by 1969", "Binary (nominal)", "Nominal",
  `smoke`, "Current smoking", "Binary (nominal)", "Nominal",
  `arcus`, "Arcus senilis", "Binary (nominal)", "Nominal",
  `dibpat`, "Behavioral pattern (A/B)", "Binary (nominal)", "Nominal",
  `behpat`, "Behavioral pattern (A1/A2/B3/B4)", "Nominal", "Nominal",
  `wghtcat`, "Weight category", "Ordinal", "Ordinal",
  `agec`, "Age group", "Ordinal", "Ordinal"
) |>
knitr::kable()
```

```

nodes <- tibble::tribble(
  ~id,    ~x,    ~y,    ~label,
  "V",    5,    4.5,  "Variables",
  "N",    2.5,  3,    "Numerical\n(quantitative)",
  "C",    7.5,  3,    "Categorical\n(qualitative)",
  "I",    1,    1.5,  "Interval\n(no true zero)\ne.g. temp. in deg C",
  "R",    4,    1.5,  "Ratio\n(true zero)\ne.g. age, weight",
  "CT",   3,    0,    "Continuous\ne.g. age, BMI",
  "CNT",  5,    0,    "Count\n(discrete)\ne.g. cigs/day",
  "NOM",  6.5,  1.5,  "Nominal\n(unordered)\ne.g. blood type",
  "ORD",  8.5,  1.5,  "Ordinal\n(ordered)\ne.g. wt. category",
  "BIN",  6.5,  0,    "Binary\n(2 categories)\ne.g. CHD event"
)

edges <- tibble::tribble(
  ~from, ~to,
  "V",   "N",
  "V",   "C",
  "N",   "I",
  "N",   "R",
  "R",   "CT",
  "R",   "CNT",
  "C",   "NOM",
  "C",   "ORD",
  "NOM", "BIN"
) |>
dplyr::left_join(
  dplyr::select(nodes, id, x, y),
  by = c("from" = "id")
) |>
dplyr::rename(x_from = x, y_from = y) |>
dplyr::left_join(
  dplyr::select(nodes, id, x, y),
  by = c("to" = "id")
) |>
dplyr::rename(x_to = x, y_to = y)
fill_colors <- c(
  "V" = "#f0f0f0",
  "N" = "#d0e8ff", "C" = "#ffe8d0",
  "I" = "#e8f4ff", "R" = "#e8f4ff",
  "CT" = "#c8e8ff", "CNT" = "#c8e8ff",
  "NOM" = "#ffe0c0", "ORD" = "#ffe0c0",
  "BIN" = "#ffd0a0"
)

ggplot2::ggplot() +
  ggplot2::aes() +
  ggplot2::geom_segment(
    data = edges,
    ggplot2::aes(
      x = x_from, y = y_from - 0.45,
      xend = x_to, yend = y_to + 0.45
    ),
    color = "grey50"
  ) +

```

Table 1: Selected WCGS variables and their types

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| Variable | Description                          | Type             | Scale   |
|----------|--------------------------------------|------------------|---------|
| age      | Age (years)                          | Continuous       | Ratio   |
| chol     | Total cholesterol                    | Continuous       | Ratio   |
| sbp      | Systolic blood pressure              | Continuous       | Ratio   |
| bmi      | Body mass index (kg/m <sup>2</sup> ) | Continuous       | Ratio   |
| weight   | Weight (lbs)                         | Continuous       | Ratio   |
| ncigs    | Cigarettes per day                   | Count (discrete) | Ratio   |
| chd69    | CHD event by 1969                    | Binary (nominal) | Nominal |
| smoke    | Current smoking                      | Binary (nominal) | Nominal |
| arcus    | Arcus senilis                        | Binary (nominal) | Nominal |
| dibpat   | Behavioral pattern (A/B)             | Binary (nominal) | Nominal |
| behpat   | Behavioral pattern (A1/A2/B3/B4)     | Nominal          | Nominal |
| wghtcat  | Weight category                      | Ordinal          | Ordinal |
| agec     | Age group                            | Ordinal          | Ordinal |

## Random variables

### Binary variables

**Definition 0.13** (Binary random variable). A **binary variable** is a random variable which has only two possible values in its range.

**Exercise 0.2** (Examples of binary variables). What are some examples of binary variables in health sciences and data science?

#### Solution

*Solution 0.2.* Examples of binary outcomes include:

- exposure (exposed vs unexposed)
- disease status (diseased vs healthy)
- recovery (recovered vs unrecovered)
- relapse (relapse vs remission)
- return to hospital (returned vs not returned)
- vital status (dead vs alive)

## Count variables

**Definition 0.14** (Count variable). A **count variable** is a random variable whose possible values are some subset of the non-negative integers; specifically, a random variable  $X$  such that:

$$\mathcal{R}(X) \subseteq \mathbb{N}$$

**Exercise 0.3** (Examples of count variables). What are some examples of count variables?

### Solution

*Solution 0.3.* Examples of count variables include:

- number of fish in a pond
- number of cyclones per season
- seconds of tooth-brushing per session (when rounded)
- infections per person-year
- visits to an emergency room per person-month
- car accidents per 1,000 miles driven

## Probability distributions for count outcomes

Standard probability distributions for count outcomes include:

- [Poisson distribution](#)
- [Negative binomial distribution](#)

## References

Dobson, Annette J, and Adrian G Barnett. 2018. *An Introduction to Generalized Linear Models*. 4th ed. CRC press. <https://doi.org/10.1201/9781315182780>.